

Design and Implementation of a Database Management System for Robotics Laboratory Equipment, Component Booking and Maintenance Management

Undergraduate project review

DATABASE MANAGEMENT SYSTEM • ROBOTICS LABORATORY OPERATIONS

A structured data model for managing laboratory equipment and components with booking, project allocation, usage tracking, maintenance, and complete history.

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What We'll Cover

01 Project Overview & Problem Statement

02 Aim & Objectives

03 Users & Roles

04 Resource & Inventory Management

05 Booking, Allocation & Lifecycle Tracking

06 Maintenance, Vendors & Calibration

07 Database & System Design

08 Timeline & Review Schedule

Project Overview

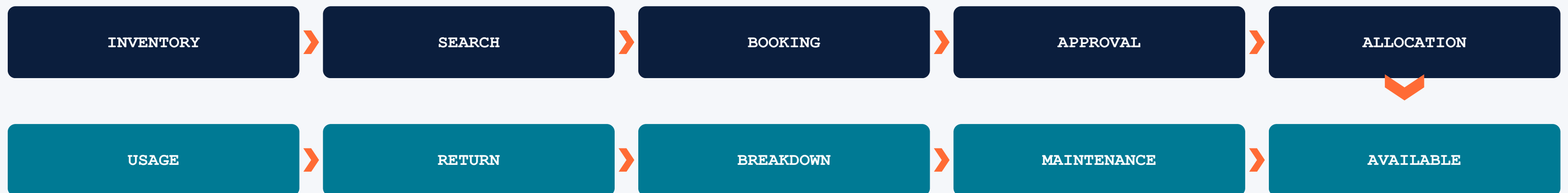
ROBO-LAB is a database-driven management system built for a college Robotics Laboratory. The lab holds many resource types – boards, sensors, motors, robotic arms, drones, 3D printers, tools and consumables – shared across students, mini-projects, final-year projects and hackathons.

Managed through registers or spreadsheets, it becomes difficult to know availability, location, current user, project, booking status, return date and maintenance history. ROBO-LAB centralizes all of this in one SQL database.

CORE IDEA

- ✓ Centralized SQL database
- ✓ Tracks the complete resource lifecycle
- ✓ Connected to a simple web application
- ✓ Full history of who, what, when & why

THE RESOURCE LIFECYCLE



The Problem With Manual Tracking

Registers and spreadsheets break down as a shared lab scales



Low Visibility

Hard to know what equipment is currently available.



Duplicate Requests

Multiple students request the same item at once.



Unknown Holder

Unclear who currently has a component, or for which project.



Late Returns

Components are not returned on time, with no reminders.



No History

No structured record of usage, breakdowns or maintenance.



Stock & Calibration Drift

Quantities go inaccurate; calibration dates get missed.

Key Objectives



Inventory & Resources

- Centralized resource database
- Categorize equipment & components
- Unique IDs & serial tracking
- Quantity & minimum stock levels



Booking & Allocation

- Search & request bookings
- Approval workflow for requests
- Allocate to students & projects
- Track issue & return dates



Maintenance & Vendors

- Breakdown reporting
- Maintenance job tracking
- Vendor & service history
- Equipment calibration tracking



Reporting & Integrity

- Complete usage & allocation history
- Notifications for key events
- Useful SQL-based reports
- Data integrity via DB constraints

Users & Roles

Four role types, each scoped to what they need to do



Admin

- Add users, roles & laboratories
- Add equipment, components & categories
- View all bookings & reports
- Manage vendors



Lab Technician

- Issue & receive equipment/components
- Record breakdowns
- Create & update maintenance jobs
- Check inventory & manage stock



Faculty

- View bookings
- Approve or reject requests
- View project usage & allocations
- View reports

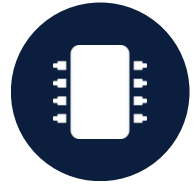


Student

- Search resources & check availability
- Request bookings, view status
- Record usage
- Request returns & report breakdowns

Laboratory Resources

Six categories cover everything the lab stocks



Controllers

Arduino UNO/Mega · ESP32 · Raspberry Pi · STM32



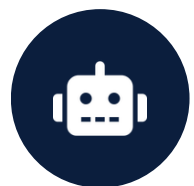
Sensors

Ultrasonic · IR · MPU6050 · DHT11 · Camera



Motors & Actuators

Servo · DC · Stepper · BLDC · Motor Driver



Robotics Equipment

Robotic Arm · Drone · 3D Printer · CNC Machine



Laboratory Tools

Multimeter · Oscilloscope · Soldering Station



Consumables

Jumper Wires · Resistors · LEDs · Breadboards

Equipment vs. Components

Two tracking models inside the same database



EQUIPMENT

Individually tracked assets, each with a unique ID

EQ001	Robotic Arm
EQ002	3D Printer
EQ003	Oscilloscope
EQ004	Drone



COMPONENTS

Tracked by quantity, drawn from shared stock

Arduino UNO	20 units
ESP32	15 units
Ultrasonic Sensor	30 units
Servo Motor	20 units
MPU6050	15 units

Note: individually valuable components can still be assigned an asset ID or serial number for tighter tracking.

Component Information & Use Cases

TRACKED FIELDS

- Component ID
- Component Name
- Category
- Manufacturer
- Model Number
- Quantity
- Available Quantity
- Minimum Stock Level
- Purchase Date
- Vendor
- Datasheet Link
- Specifications
- Description
- Current Status

MPU6050

Sensor

Qty 15 · Available 8 · Min 5

Use: Accelerometer & gyroscope measurements

Ultrasonic Sensor

Sensor

Qty 30 · Available 19

Use: Distance measurement & obstacle detection

ESP32

Controller

Qty 15 · Available 9

Use: IoT & wireless robotics applications

Servo Motor

Actuator

Qty 20 · Available 14

Use: Position control & robotic movement



Student Projects & Component Allocation



P001 — Autonomous Line Following Robot

LEAD

Student A

EXPECTED END

30-09-2026

START

01-09-2026

STATUS

Active

ALLOCATED COMPONENTS

Arduino UNO	1 unit
Ultrasonic Sensor	2 units
Servo Motor	2 units
DC Motor	2 units
Motor Driver	1 unit



Equipment Booking & Conflict Prevention



Booking Request

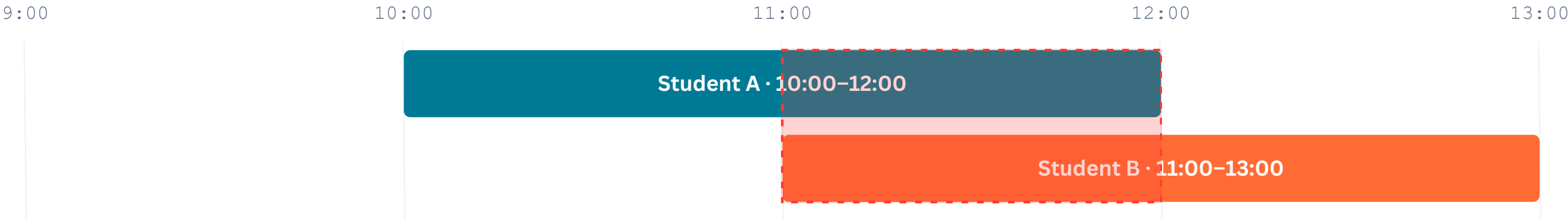
Equipment: **Robotic Arm**
Date / Time: **10-09-2026, 10:00–12:00**
Purpose: **Pick & Place Robot Project**
Requested By: **Student A**

STATUS FLOW



Faculty / Admin approve or reject each request.

BOOKING CONFLICT PREVENTION — ROBOTIC ARM



Overlap — the second request cannot also be approved.

The same rule applies across every equipment item in the lab — approval always checks existing bookings first.

Equipment Status & Current-User Tracking

POSSIBLE EQUIPMENT STATES

Available

In Use

Broken

Booked

Under Maintenance

Retired

CURRENT USER LOOKUP



Robotic Arm

EQ001

CURRENT USER

Student A

PROJECT

Pick and Place Robot

ISSUED

05-09-2026

EXPECTED RETURN

15-09-2026

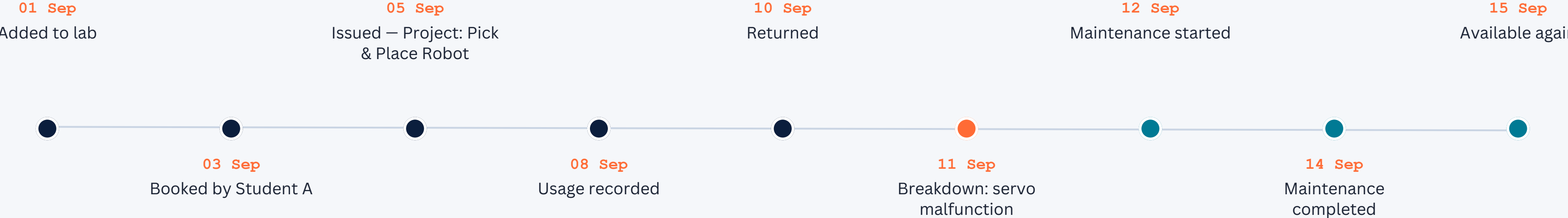
STATUS

In Use

Every equipment record always answers two questions at a glance: what state is it in, and who is responsible for it right now.

Complete Resource History — Example Lifecycle

EQ001 · Robotic Arm



This event trail — add, book, issue, use, return, breakdown, repair, available — is the asset lifecycle behind every resource in ROBO-LAB.

Vendor Management & Calibration



Vendor Profile Example

Vendor: **ABC Robotics Solutions**

Contact: **XYZ**

Phone: **XXXXXXXXXX**

Email: **example@email.com**

Service: **Robotic arm repair & calibration**

Vendor history answers: which vendor serviced an item, how many times, at what cost, and when.



Calibration Tracking

- Equipment reference
- Calibration Date
- Next Calibration Date
- Calibration Agency
- Certificate Number
- Remarks

A Calibration-Due report flags multimeters, oscilloscopes, sensors and robotic equipment approaching their next due date.

Notifications & Search



Notifications

Booking Approved

Return Due Tomorrow

Overdue Return

Calibration Due

Maintenance Completed

Breakdown Reported

Low Component Stock

"Arduino UNO allocated to you is due for return tomorrow."



Smart Search

Searchable by: Name · Category · Asset ID · Serial No. · Lab · Current User · Project · Vendor · Status



Arduino

Arduino UNO #001

Available

Arduino UNO #002

In Use · P001

Arduino Mega #001

Under Maintenance

SQL-Based Reports

Turning transactional data into decisions



Equipment Availability



Current Allocation



Student Usage



Project Resource



Overdue Return



Breakdown Report



Maintenance History



Downtime Report



Calibration Due



Low Stock



Vendor Service



Utilization



Database Design -- Core Tables

18 relational tables organized into three functional groups

MASTER DATA

ROLES
USERS
LABORATORIES
EQUIPMENT_CATEGORIES
EQUIPMENT
COMPONENT_CATEGORIES

TRANSACTIONS

COMPONENTS
PROJECTS
PROJECT_MEMBERS
BOOKINGS
EQUIPMENT_ALLOCATION
USAGE_LOGS

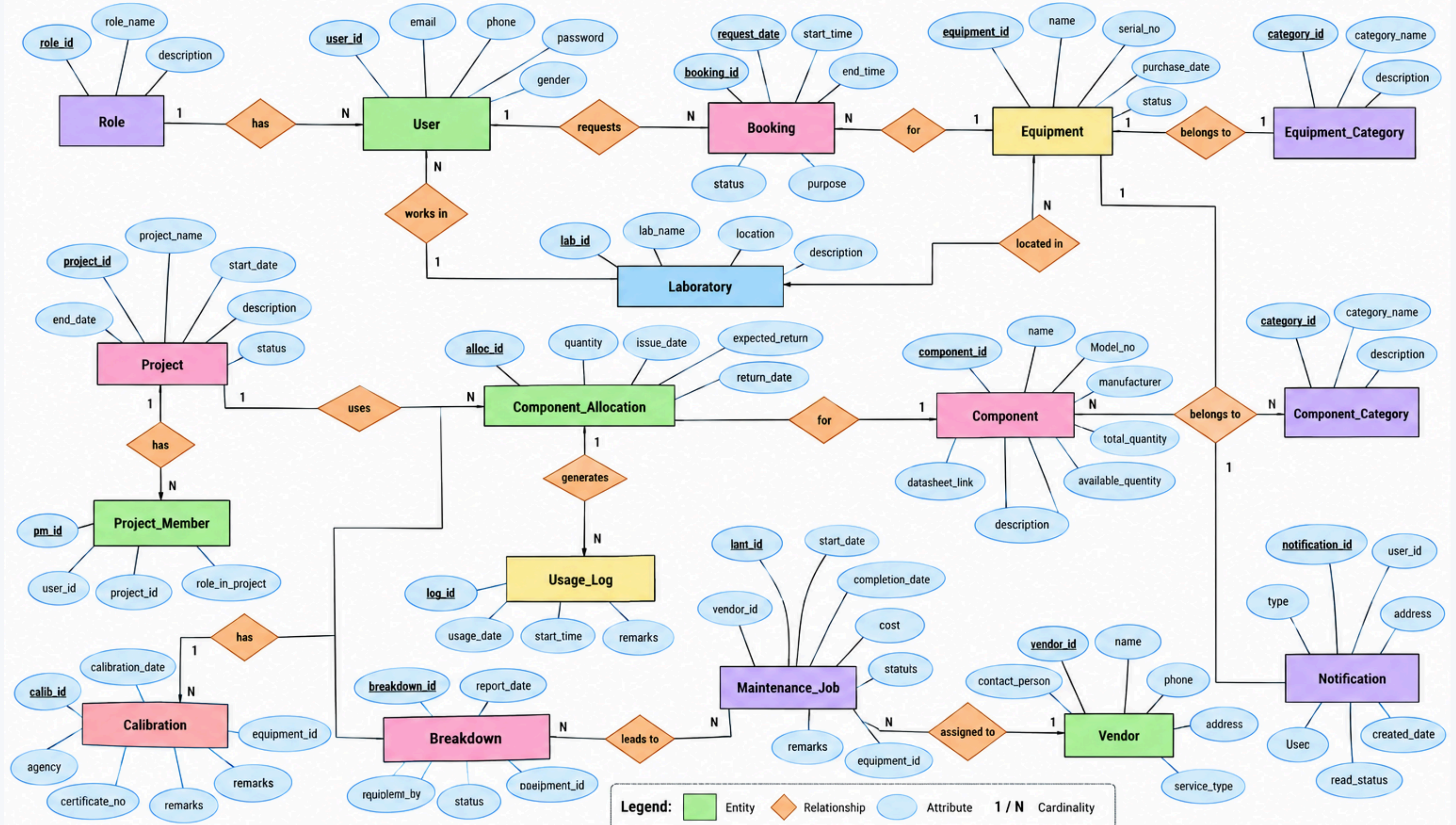
SUPPORT & HISTORY

RETURNS
BREAKDOWNS
MAINTENANCE_JOBS
VENDORS
CALIBRATIONS
NOTIFICATIONS

Optional add-ons if consumables need explicit tracking: CONSUMABLES, CONSUMABLE_ISSUES

ROBO-LAB

Robotics Laboratory Equipment, Component Booking & Maintenance Management System ER Diagram (Conceptual)



System Architecture & Tech Stack



Frontend

HTML · CSS · JavaScript



Backend

Python Flask



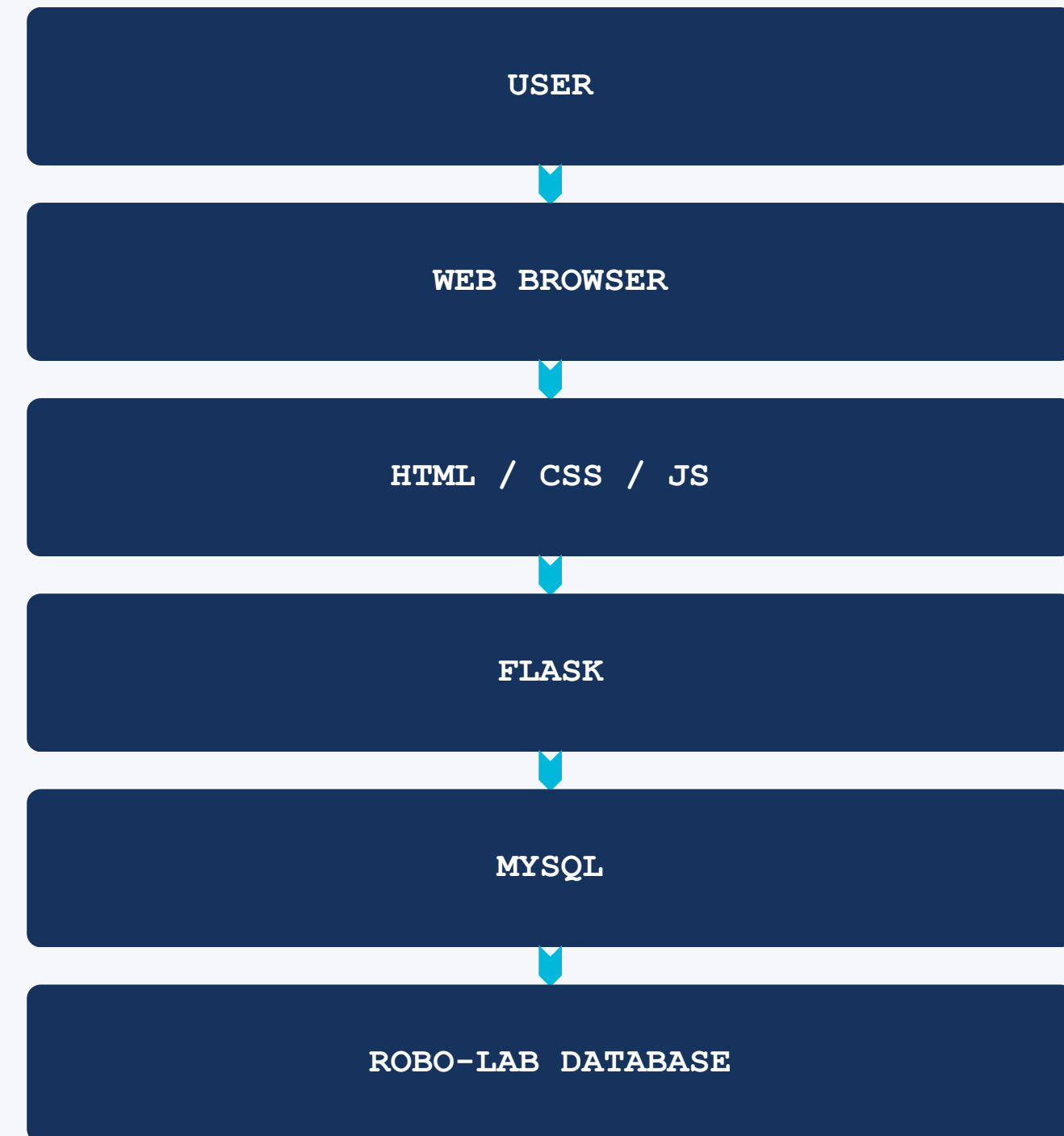
Database

MySQL



Connector

MySQL Connector / Python



Dashboard & Application Pages

ROBO-LAB DASHBOARD

50

Total Equipment

32

Available

8

In Use

6

Under Maintenance

2

Broken

2

Overdue

RECENT ACTIVITY

• Arduino UNO issued to Student A

• Robotic Arm booked for 10-09-2026

• 3D Printer maintenance completed

• ESP32 returned in good condition

MAIN APPLICATION PAGES

Login	Dashboard	Equipment
Components	Projects	Users
Bookings	Allocations	Usage
Returns	Breakdowns	Maintenance
Calibration	Vendors	Notifications
Reports		

Breakdown → Maintenance → Resolution



Breakdown Report Example

Equipment:	3D Printer
Problem:	Nozzle blockage
Severity:	Medium
Reported By:	Student B
Date:	08-09-2026
Status:	Open

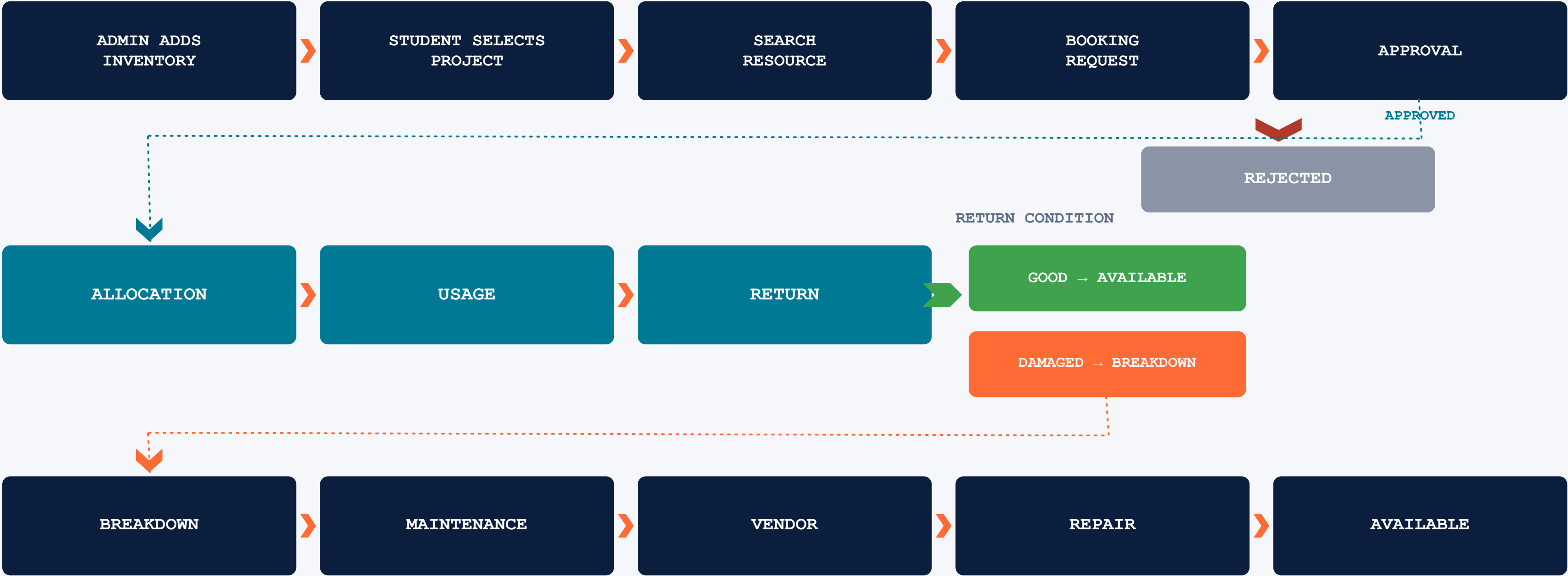


Maintenance Job Fields

- Maintenance ID & Equipment
- Linked Breakdown Record
- Technician & Vendor
- Start / Completion Date
- Cost & Parts Replaced
- Status & Remarks

Complete System Workflow

From inventory setup to repair and back to available – the heart of ROBO-LAB





Thank You



Questions & Discussion – ROBO-LAB

